

Unitree Week 34

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This Week

- Research on more Unitree Physical Setup
- Research on Reinforcement Learning Development

Online Unitree G1 Humanoid Setup and Training Guide

This playlist provides information on physical Unitree G1 setup, which could be useful for when setting up the Unitree R1. It contains information such as:

- The G1 Training Mode
- Updating the G1
- Fold and packing the G1
- Binding the Controller to the G1
- Connecting and Startup for the G1
- G1 Posture and Movement Controls
- G1 Calibration

The Unitree R1 obtains a similar training guide provided by Unitree itself.

Note on Online Unitree R1 Humanoid Setup and Training Guide

Note: “Wrist-mounted RGB cameras and an optional 3D LiDAR sensing kit for advanced mapping and tracking“

Custom Reinforcement Learning Setup Guide

Important factors in designing any behavior for reinforcement learning in humanoid robotics consists of:

- What does the robot observe?
- What do you tell the robot to accomplish?
- What actions does the robot take?
- Factors that measures success and failure (and the mathematical function that equates to this)
- When does an episode end?

Custom Reinforcement Learning Setup Guide

Observations:

- Joint Positions
- Joint Velocities
- IMU
- Robot Orientation
- Target Information

Goal

- Ex: “reach this point”

Actions:

- Joint target positions
- High-level velocity commands

Rewards:

- Numeric Measurement of Success

Termination:

- Falling
- Collision
- Time Limit
- Success

Example: Locomotion

In order to train the R1 EDU to do locomotion in the flat environment, we train it by:

- Command/goal: $[v_x^*, v_y^*, \text{yaw-rate}^*]$
- Observations:
 - IMU angular velocity
 - Orientation
 - Desired velocity command
 - Walking phase
 - Joint positions and velocities
 - Previous Actions

Example: Locomotion

- Actions:
 - Desired position for every controlled R1 joint
- Rewards:
 - Follow velocity command and turning command
 - Use alternating gaits.
 - Stay near good posture
 - Minimizing factors: tilting, sliding, hard impact, joint acceleration, sporadic actions, self collisions, excessive body movements/
- Termination
 - 20s episode finished or
 - Robot tips beyond 70 degrees.

Locomotion Reward Terms

Reward Term	Weight	Purpose
Linear Velocity Tracking	+1.0	Move at request v_x, v_y
Angular Velocity Tracking	+1.0	Turn at requested Yaw Rate
Pose	+1.0	Maintain a sensible nominal posture
Gait	+0.5	Alternate feet according to gait phase
Torso Orientation	-1.0	Avoid leaning/falling
Torso Angular Moment	-0.05	Avoid sporadic roll/pitch rotations
Angular Moment	-0.025	Avoid Excessive whole-body flailing
Termination	-200	High penalty for falling/terminal failure

Locomotion Reward Terms (continued)

Reward Term	Weight	Purpose
Joint Acceleration	-2.5e-7	Avoid high joint acceleration
Joint Limits	-10	Avoid exceeding joint limits
Action Rate	-0.05	Avoid abruptly changing commands
Foot Clearance	-1.0	Lift feet to sensible height
Foot Slip	-0.25	Don't slide planted feet
Landing Impact	-0.001	Don't slam feet down
Stand Still	-1.0	When command is zero, maintain standing pose
R1 Self-Collision	-1.0	No contact with it's own body

References

- <https://www.youtube.com/playlist?list=PLUZOCha5G72apYRFWvUpbi44AUM8S731l>
- https://support.unitree.com/home/en/R1_developer/Operational_guidance
- https://github.com/unitreerobotics/unitree_rl_mjlab/blob/main/src/tasks/velocity/config/r1/env_cfgs.py
- https://github.com/unitreerobotics/unitree_rl_mjlab/blob/main/src/tasks/velocity/velocity_env_cfg.py